

Software Requirements Specification
for
Student CGPA Management System

Version 1.0

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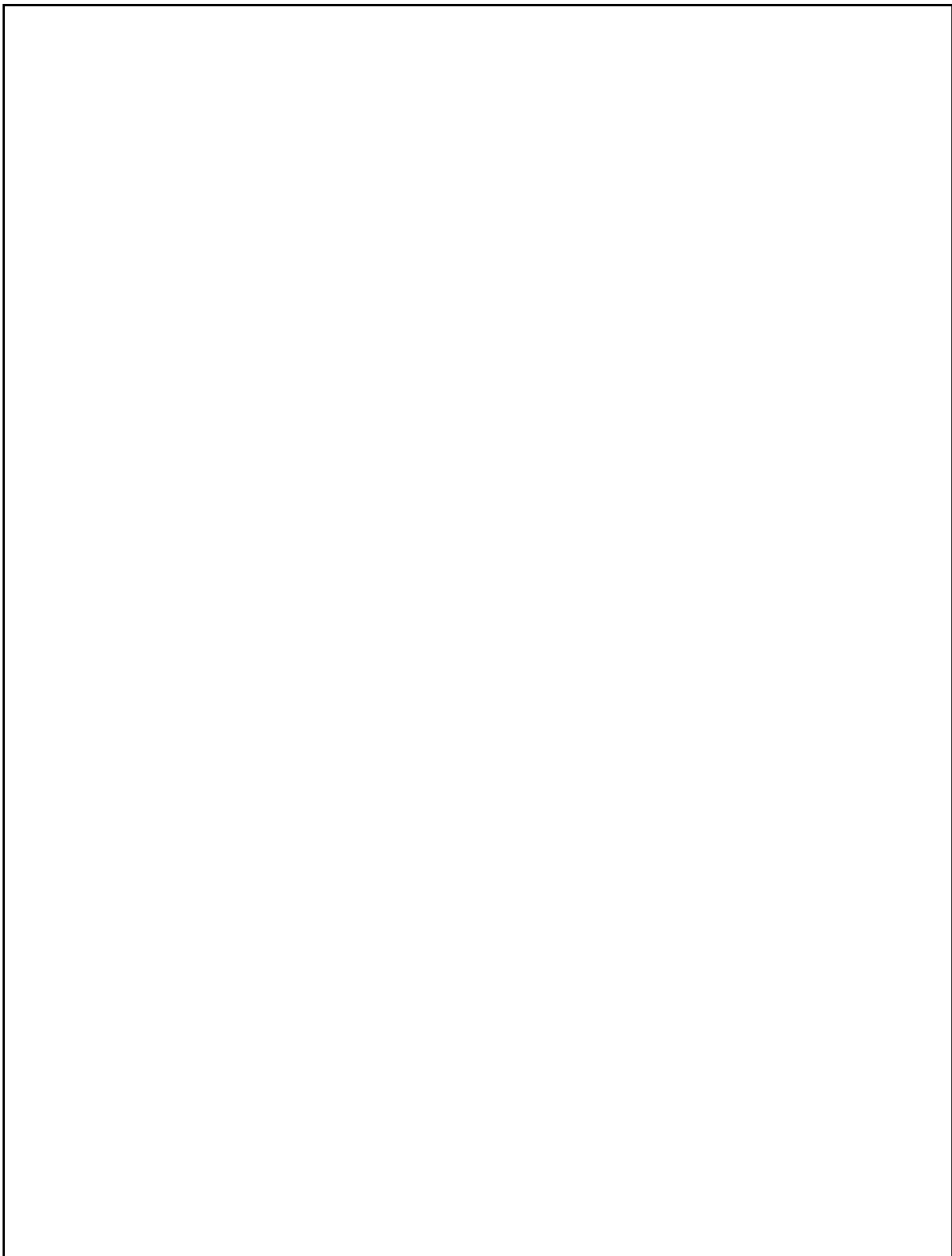


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1. Introduction

1.1 Purpose

This document specifies the software requirements for Version 1.0 of the Student CGPA Management System, a console-based application developed in the C programming language. It describes the functional and non-functional requirements for a system that records a student's academic performance across multiple semesters and automatically computes semester-wise SGPA and an overall credit-weighted CGPA. This SRS is intended for academic evaluation as part of a mini project submission.

The main purpose of the Student CGPA Management System is to simplify the process of maintaining and evaluating a student's academic performance. The system reduces manual calculation errors by automatically converting grades into grade points and calculating SGPA and CGPA. It provides a structured way to enter semester-wise subject information and view the student's overall academic performance.

The application is designed to provide a simple and user-friendly console-based environment where the user can enter academic details and obtain calculated results without performing the calculations manually.

1.2 Document Conventions

Requirement statements use the word 'shall' to indicate a mandatory requirement. Functional requirements are grouped and labeled FR-1, FR-2, and so on. Section numbering follows the standard IEEE/SRS convention used throughout this document.

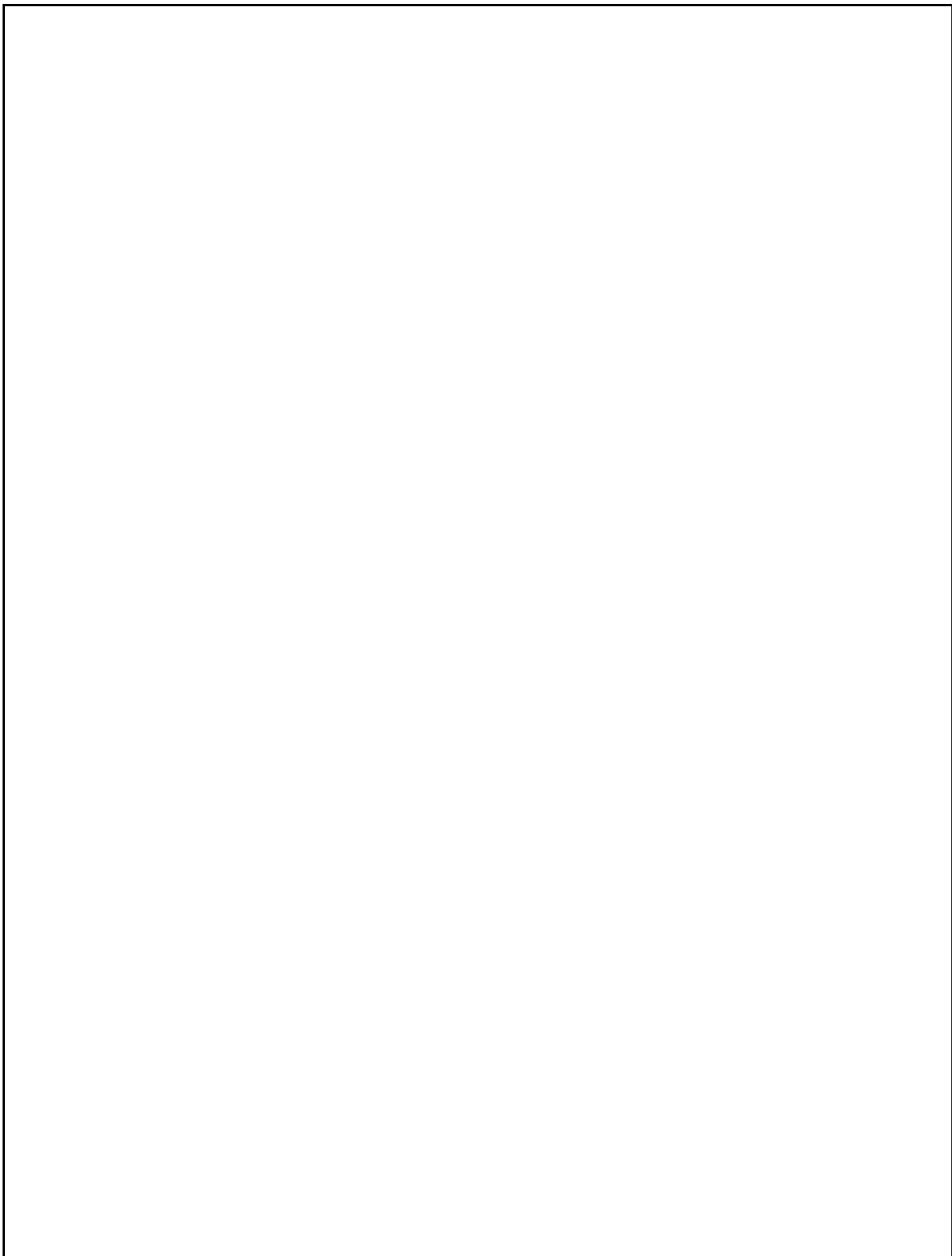
1.3 Intended Audience and Reading Suggestions

This document is intended for the course instructor/evaluator, and for the student developer as a reference for implementation and testing. Section 2 gives a high-level overview suitable for a first read; Sections 3-5 provide detailed requirements useful for implementation and verification.

1.4 Product Scope

The Student CGPA Management System is a single-user, menu-driven console application that:

- Captures a student's personal and academic identity (name, registration number, number of completed semesters).
- Records subject-wise details (name, credit, grade) for each completed semester.
- Automatically converts letter grades into grade points using a fixed grading scale.
- Computes the Semester Grade Point Average (SGPA) for each semester.



- Computes the overall Cumulative Grade Point Average (CGPA) using credit-weighted aggregation across all completed semesters.
- Displays a consolidated academic report on demand.

The system does not provide persistent storage; all data exists only in memory for the duration of a single program execution and is lost on exit.

1.5 References

- IEEE Std 830-1998, IEEE Recommended Practice for Software Requirements Specifications.
- Source code: cgpa_management.c (C, standard library only).

2. Overall Description

2.1 Product Perspective

The system is a new, standalone, self-contained console application with no external dependencies, database, or network connectivity. It uses in-memory C structures (Student, Semester, Subject) to hold all data for a single student session.

2.2 Product Functions

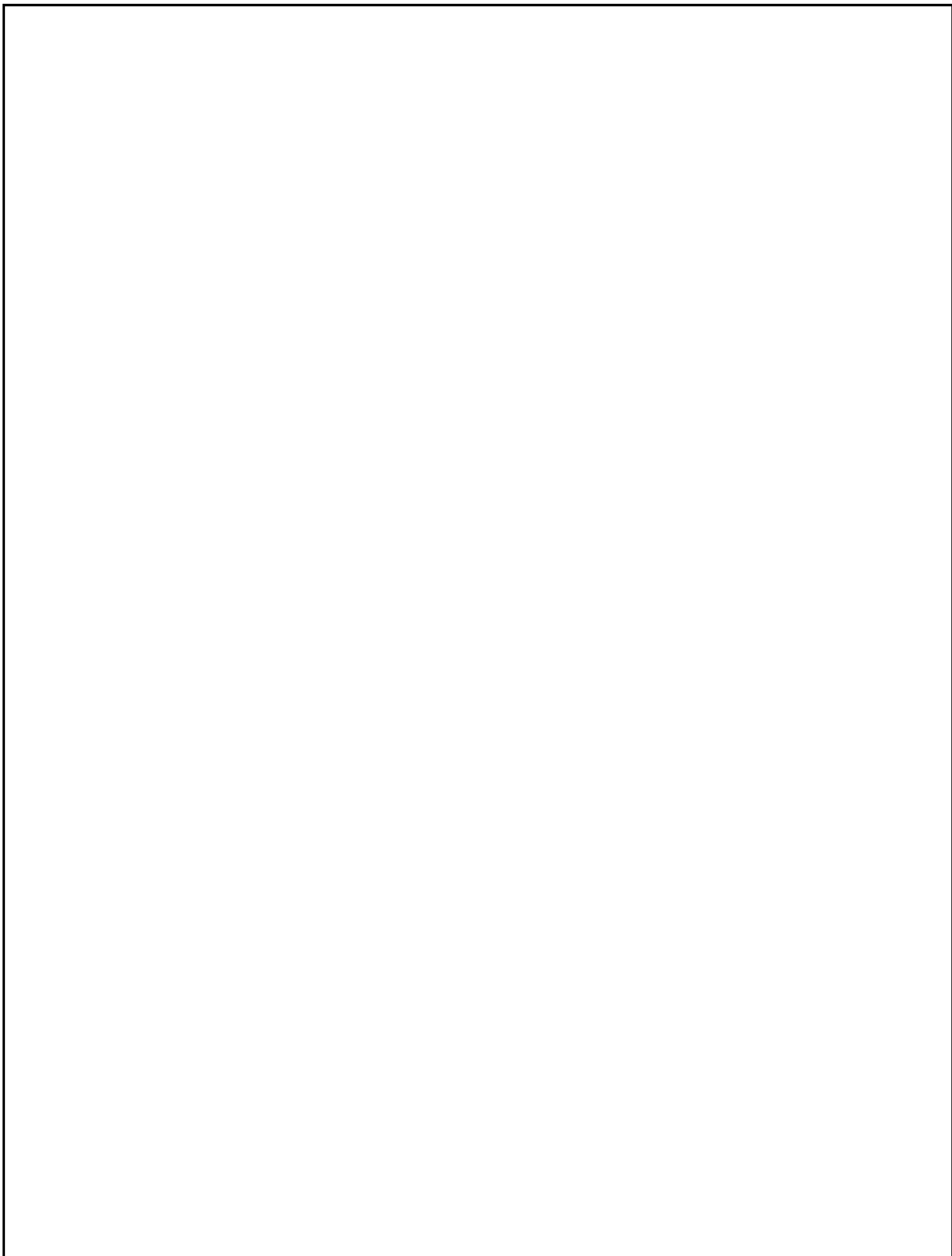
At a high level, the system provides four functions, exposed through a text menu:

	Menu Option	Function
1	Enter Student Details	Capture student name, registration number, and number of completed semesters (1-8).
2	Enter Semester Details	Capture subject count, name, credit, and grade for each subject in every completed semester; compute SGPA.
3	Display Report	Print a consolidated report of all semesters and the overall CGPA.
4	Exit	Terminate the program.

The major functions of the system are described below:

Student Detail Entry:

The system collects basic student information such as name, registration number and the number of completed semesters. This information is used to identify the student throughout the academic report.



Semester and Subject Detail Entry:

For every completed semester, the user enters the number of subjects and the details of each subject. The details include subject name, credit value and letter grade.

Grade Conversion:

The system automatically converts each letter grade into its corresponding grade point according to the predefined grading scale. This eliminates the need for the user to manually calculate grade points.

SGPA Calculation:

The system calculates the SGPA of each semester using the credit value and grade point of every subject.

CGPA Calculation:

The system calculates the overall CGPA by combining the academic performance of all completed semesters using the total credits and grade points.

Report Display:

The system displays the student's academic information, subject-wise details, semester-wise SGPA and final CGPA in a consolidated report.

Menu Navigation:

The user can select the required operation from the main menu and continue using the system until the Exit option is selected.

2.3 User Classes and Characteristics

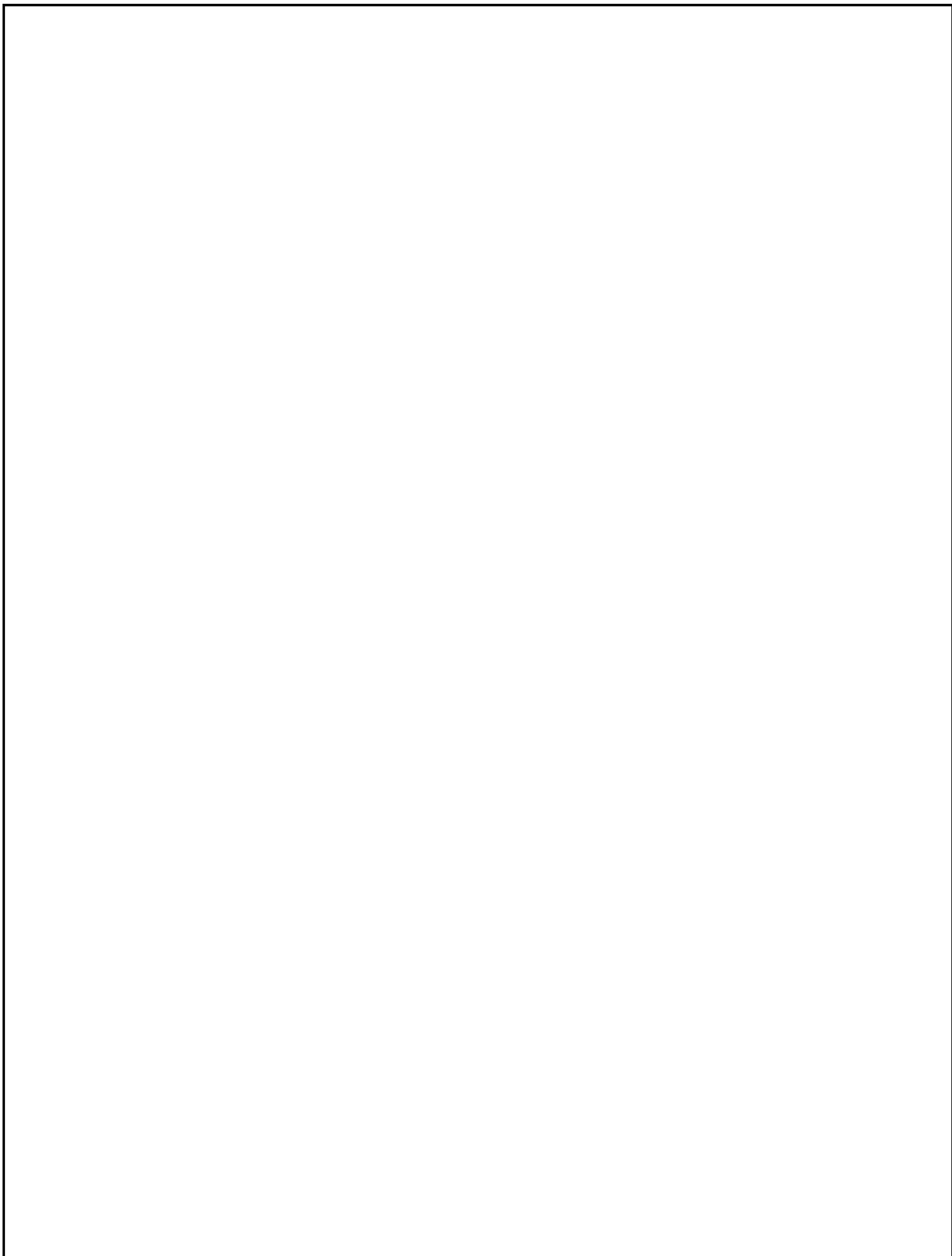
The intended user is a single student or academic staff member who is comfortable with basic keyboard-driven, text-menu interaction. No specialized training is required. There is only one user class; the system does not distinguish between administrator and regular user roles.

2.4 Operating Environment

The system runs as a console/terminal application on any platform with a standard C compiler and runtime (e.g., Windows, Linux, macOS). It has no GUI, network, or database requirements.

2.5 Design and Implementation Constraints

- Maximum of 8 semesters ($\text{MAX_SEM} = 8$).
- Maximum of 10 subjects per semester ($\text{MAX_SUB} = 10$).
- Grading scale is fixed and hard-coded (not configurable at runtime).



- No file or database persistence; data is lost when the program exits.
- Single-user, single-session console application (no concurrency or multi-user support).
- Implemented in standard C using only the `stdio.h` and `string.h` libraries.

2.6 User Documentation

On-screen menu prompts and labeled output serve as the primary user guidance. No separate user manual is provided with this version; this SRS document and in-application prompts are the available documentation.

2.7 Assumptions and Dependencies

- The user enters numeric inputs (credits, semester counts) within valid ranges.
- Grades are entered using the exact case-sensitive strings recognized by the system (O, A+, A, B+, B, C); any other value is treated as a fail grade (0 grade points).
- The program is compiled and run in a standard C runtime environment (e.g., GCC).

3. External Interface Requirements

3.1 User Interfaces

The system provides a text-based menu with four options, presented in a loop until the user selects Exit. Prompts request input via standard input (keyboard) and output is written to standard output (console). No graphical interface is provided.

The application provides a simple console-based interface. The main menu is displayed to the user whenever the program starts and after completion of an operation.

The interface provides options for entering student details, entering semester details, displaying the academic report and exiting the application. Input is provided through the keyboard and the corresponding results are displayed on the console.

The system uses clear prompts such as student name, registration number, semester number, subject name, credit and grade to guide the user during data entry. Appropriate messages are displayed for invalid menu selections.

3.2 Hardware Interfaces

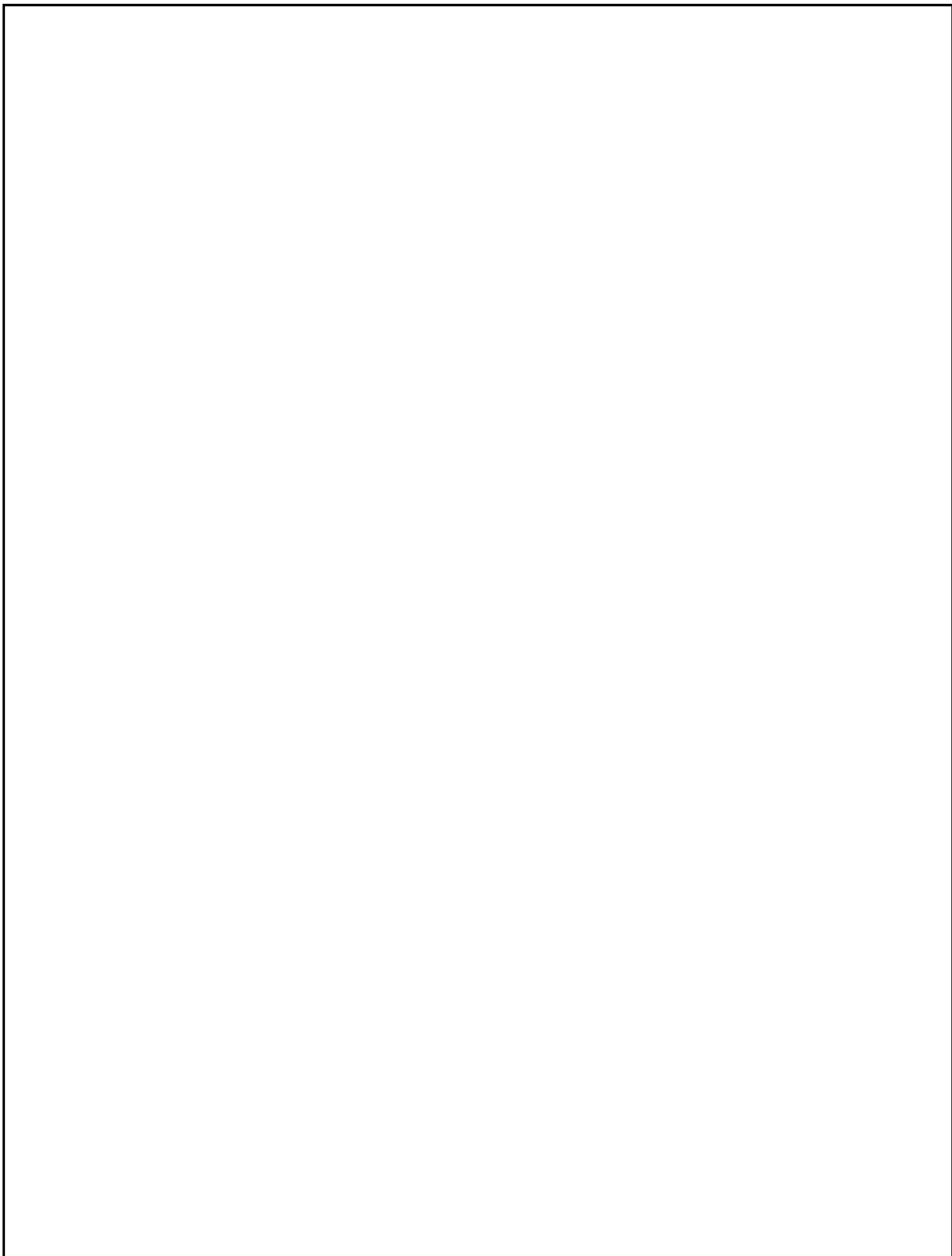
None beyond a standard keyboard and display/terminal. The application has no requirements on specific hardware devices or peripherals.

3.3 Software Interfaces

None. The program relies only on the C standard library (`stdio.h`, `string.h`) and does not interface with any external software, database, or operating-system service beyond standard I/O.

3.4 Communications Interfaces

None. The system does not perform any network or inter-process communication.



4. System Features

4.1 System Feature 1: Student Detail Entry

4.1.1 Description and Priority

Allows the user to enter the student's identifying information.

Priority: High (required before other features can be used meaningfully).

4.1.2 Functional Requirements

- FR-1.1: The system shall prompt for and accept the student's full name (including spaces).
- FR-1.2: The system shall prompt for and accept a registration number as a single token (no spaces).
- FR-1.3: The system shall prompt for and accept the number of completed semesters, an integer between 1 and 8.

4.2 System Feature 2: Semester and Subject Detail Entry

4.2.1 Description and Priority

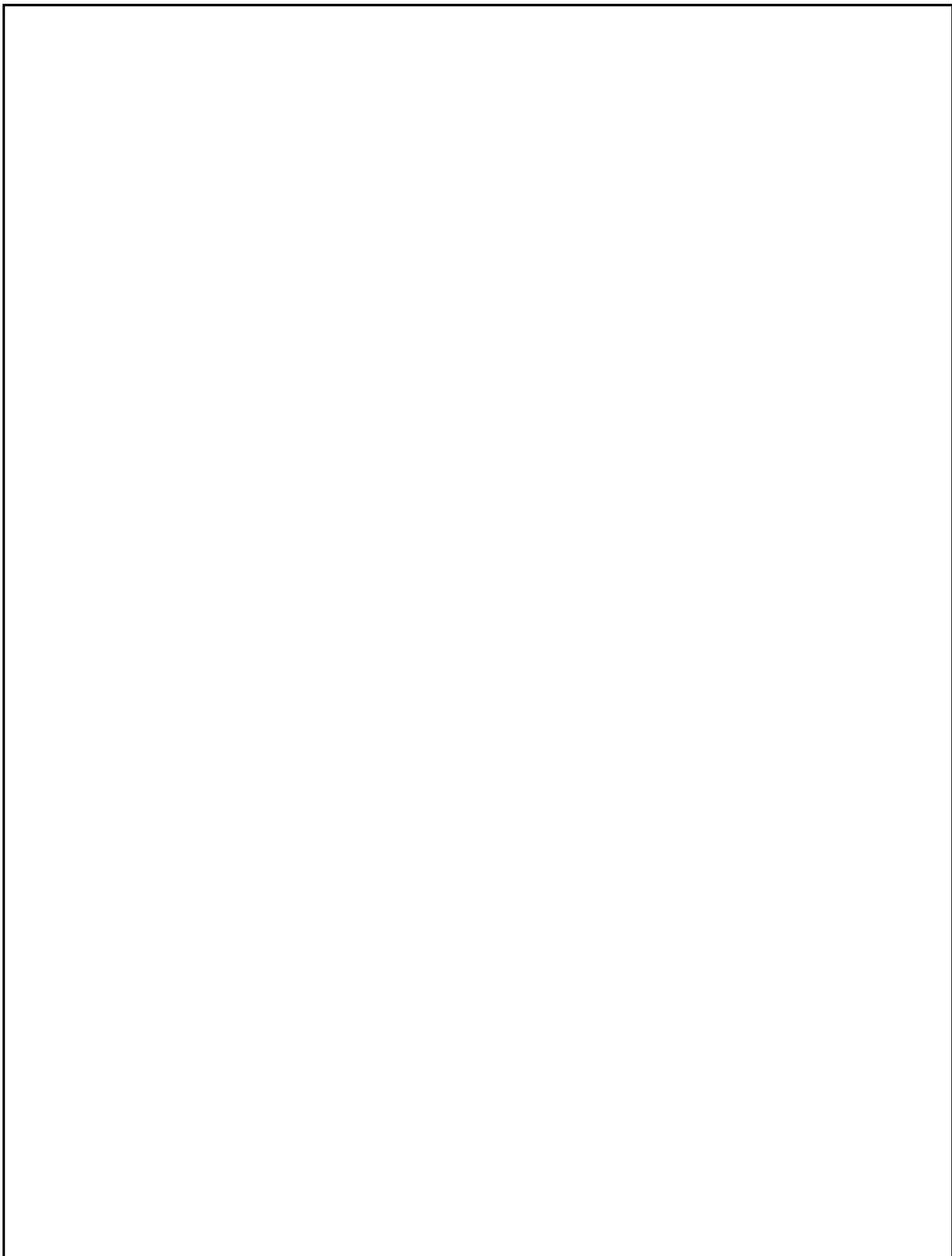
Allows the user to record subject-wise academic data for each completed semester.

Priority: High.

4.2.2 Functional Requirements

- FR-2.1: For each completed semester, the system shall prompt for the number of subjects (up to 10).
- FR-2.2: For each subject, the system shall accept the subject name, credit value (decimal), and letter grade.
- FR-2.3: The system shall convert each entered grade into a numeric grade point using the fixed grading scale below.

Grade	Grade Point
O	10
A+	9
A	8
B+	7
B	6
C	5
Any other value (e.g., U/F)	0



4.3 System Feature 3: SGPA Calculation

4.3.1 Description and Priority

Computes the Semester Grade Point Average automatically after subject data entry.

Priority: High.

4.3.2 Functional Requirements

- FR-3.1: The system shall compute total credits and total grade points (credit x grade point, summed) for each semester.
- FR-3.2: The system shall compute SGPA as total grade points divided by total credits for that semester.
- FR-3.3: If total credits for a semester is zero, the system shall set SGPA to zero for that semester to avoid division by zero.
- FR-3.4: The system shall display the computed SGPA immediately after each semester's data entry is completed.

For each semester, the system calculates the weighted grade points for every subject by multiplying the subject's credit value by its corresponding grade point.

The total weighted grade points are obtained by adding the weighted grade points of all subjects in that semester. The SGPA is then calculated by dividing the total weighted grade points by the total number of credits.

For example, if a semester contains three subjects with credits 4, 3 and 2 and corresponding grade points 9, 8 and 10, the system calculates the weighted grade points for each subject and uses their total to determine the semester SGPA.

This calculation is performed automatically after the subject details for a semester have been entered.

4.4 System Feature 4: CGPA Calculation

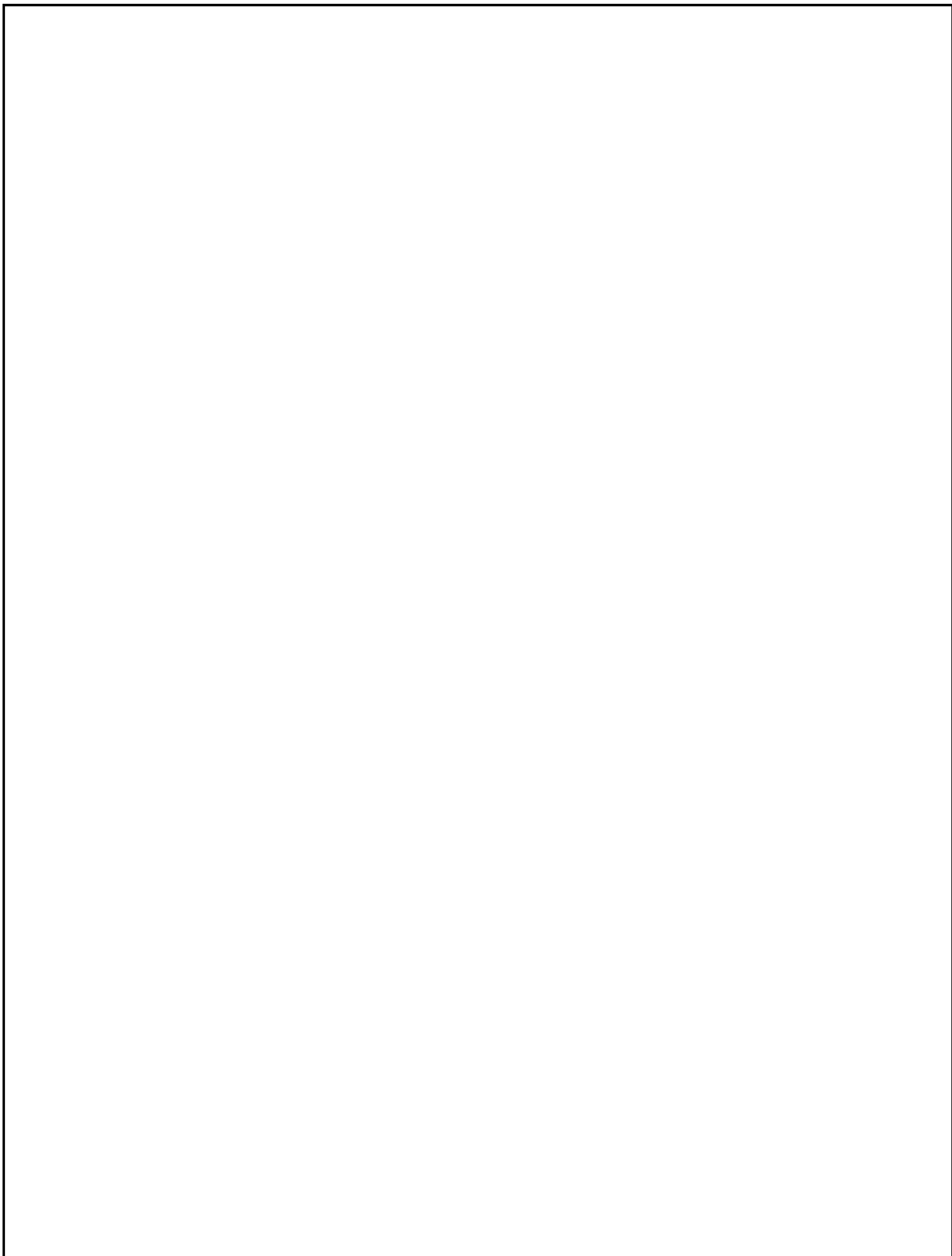
4.4.1 Description and Priority

Computes the overall credit-weighted CGPA across all completed semesters.

Priority: High.

4.4.2 Functional Requirements

- FR-4.1: The system shall aggregate total credits and total grade points across all completed semesters.
- FR-4.2: The system shall compute the overall CGPA as a credit-weighted average: sum of grade points across all semesters divided by sum of credits across all semesters.



- FR-4.3: If total aggregate credits is zero, the system shall return a CGPA of zero.

The overall CGPA represents the student's academic performance across all completed semesters. The system combines the total credits and total weighted grade points from every completed semester.

The CGPA is calculated using a credit-weighted method rather than taking a simple average of the semester SGPA's. This ensures that subjects or semesters with higher credit values have a proportionally greater contribution to the final CGPA.

The calculated CGPA is displayed as part of the final academic report.

4.5 System Feature 5: Report Display

4.5.1 Description and Priority

Displays a consolidated report of all entered data and computed results.

Priority: Medium.

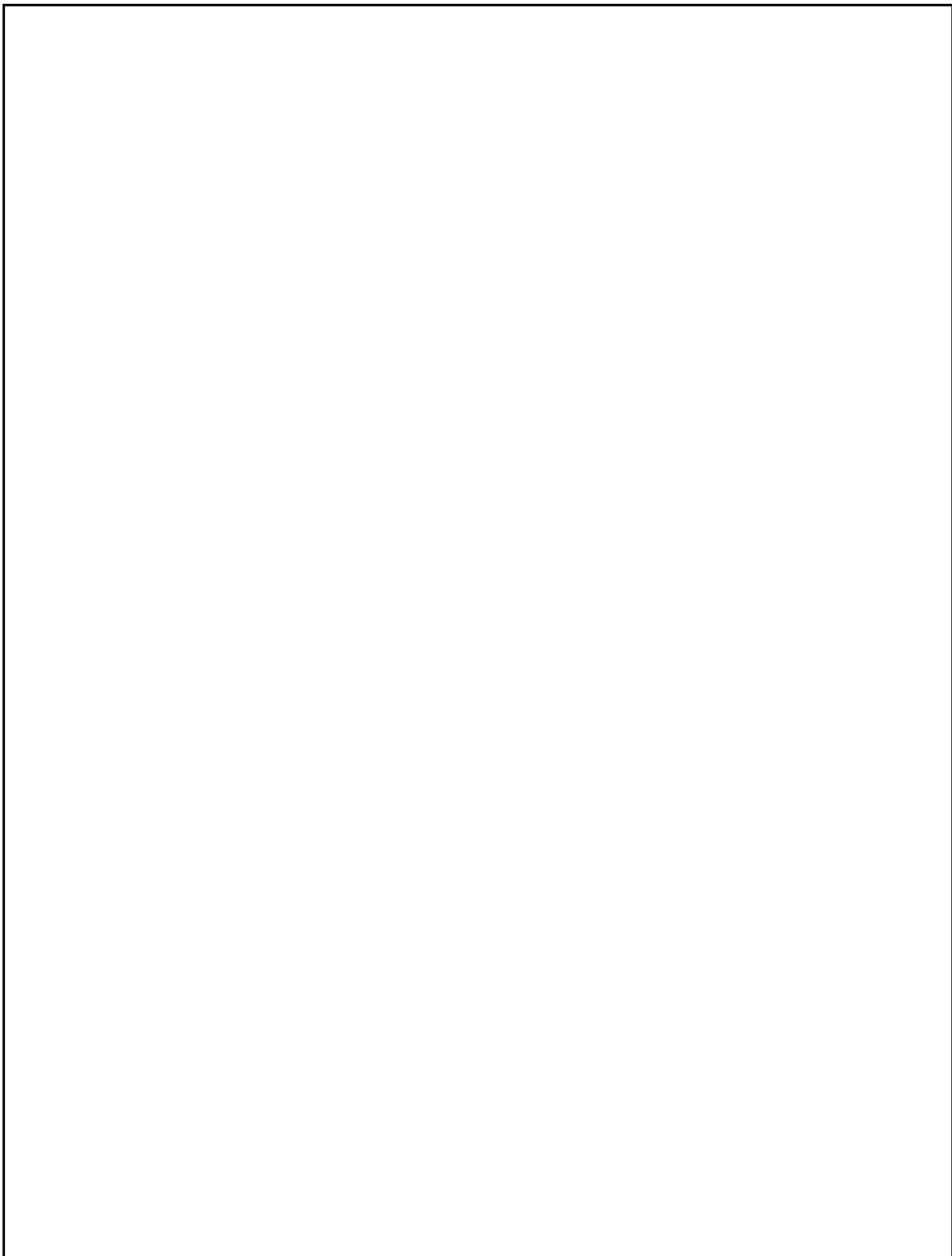
4.5.2 Functional Requirements

- FR-5.1: The system shall display the student's name and registration number.
- FR-5.2: For each of the 8 possible semesters, the system shall display subject-wise details (name, credit, grade, grade point), total credits, total grade points, and SGPA if the semester was completed, or a 'Not Completed' indicator otherwise.
- FR-5.3: The system shall display the final credit-weighted CGPA at the end of the report.

The academic report provides a consolidated view of the student's performance. It includes the student's name and registration number followed by semester-wise academic information.

For each completed semester, the report displays the subjects, corresponding credits, entered grades, calculated grade points, total credits, total weighted grade points and SGPA. Semesters that have not been completed are clearly indicated.

At the end of the report, the system displays the overall credit-weighted CGPA, allowing the user to easily review the student's complete academic performance.



4.6 System Feature 6: Menu Navigation

4.6.1 Description and Priority

Provides the main control loop and option selection for the application.

Priority: High.

4.6.2 Functional Requirements

- FR-6.1: The system shall present a repeating menu with options to enter student details, enter semester details, display the report, and exit.
- FR-6.2: The system shall display an 'Invalid Choice' message for any menu input outside the range 1-4.
- FR-6.3: The system shall terminate cleanly when option 4 (Exit) is selected.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

All calculations shall complete instantly (negligible processing time) given the small, bounded data size (maximum 8 semesters x 10 subjects). No specific throughput or response-time targets apply beyond immediate console responsiveness.

5.2 Safety Requirements

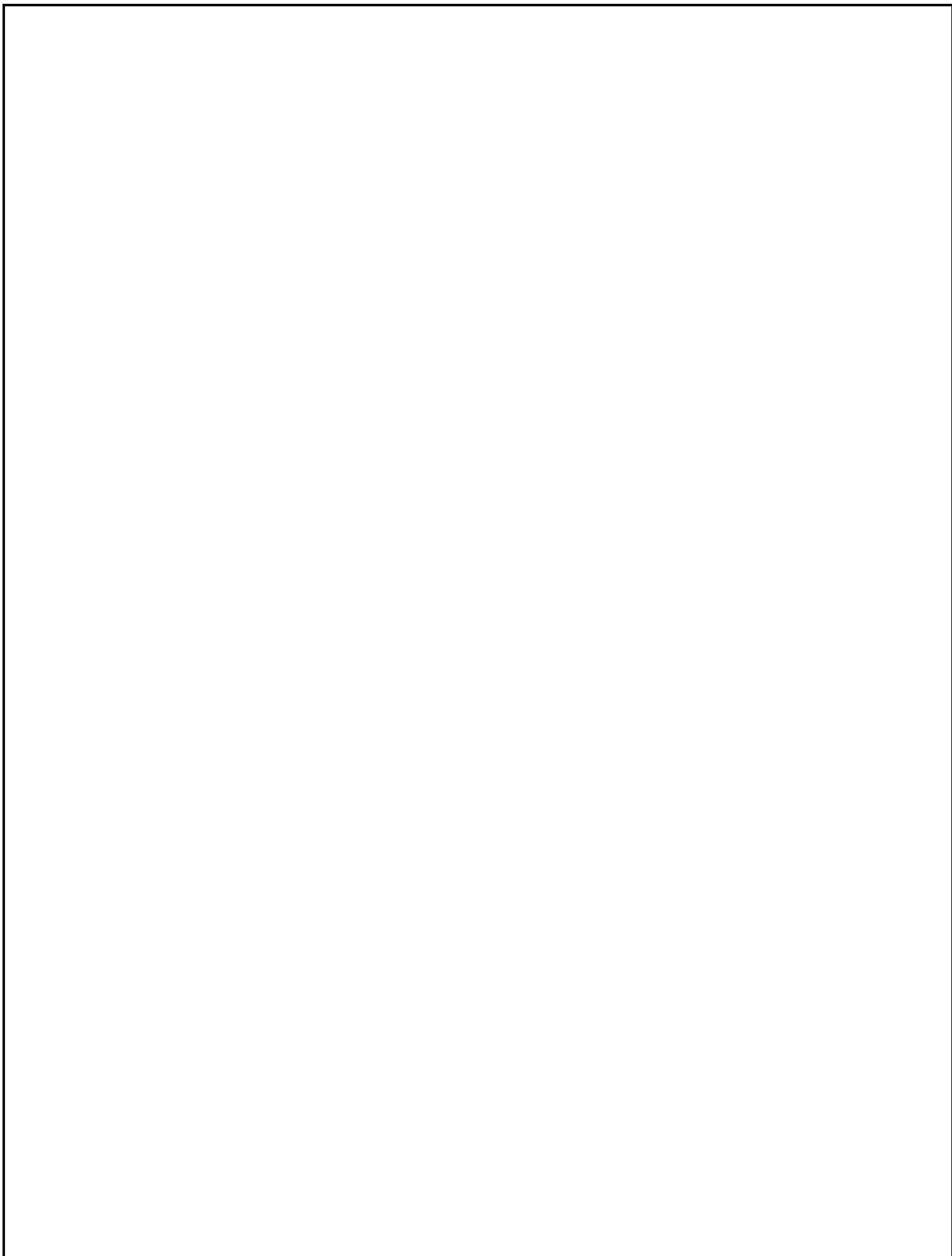
Not applicable. The system does not control any physical process or safety-critical function.

5.3 Security Requirements

Not applicable in the current version. The system has no authentication, authorization, or data-protection requirements, as it is a single-user, single-session, in-memory application with no persistent or shared data store.

5.4 Software Quality Attributes

Attribute	Requirement
Usability	Menu prompts and output labels shall be clear and self-explanatory for a first-time user.
Reliability	The system shall handle a semester or aggregate credit total of zero without crashing (no division by zero).
Portability	The system shall compile and run on any standard C compiler/runtime without modification.
Maintainability	Data is organized in modular structures (Student, Semester, Subject) to simplify future extension, e.g., adding file persistence.



5.5 Business Rules

- A maximum of 8 semesters and 10 subjects per semester may be recorded, per the constraints in Section 2.5.
- Only the six defined letter grades (O, A+, A, B+, B, C) earn non-zero grade points; any other grade string is treated as a fail (0 grade points).
- CGPA is always calculated as a credit-weighted average, never a simple average of SGPAs, so semesters or subjects with higher credit loads have proportionally greater influence on the final CGPA.